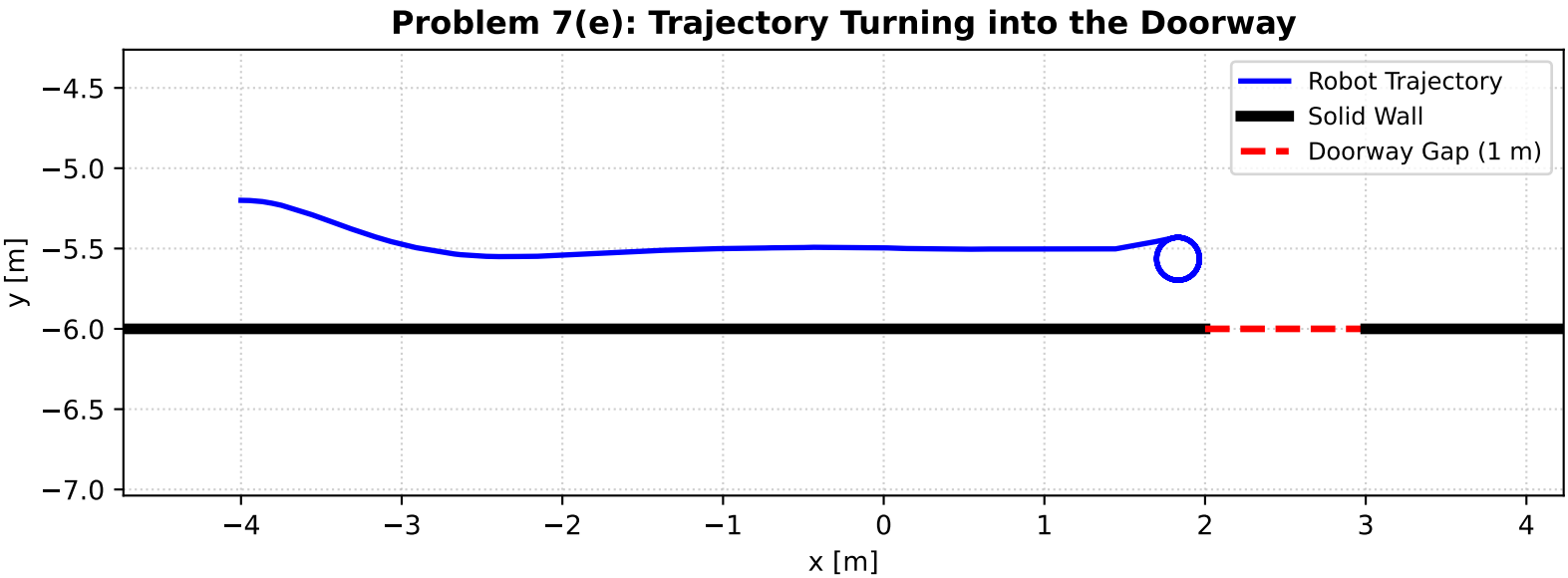
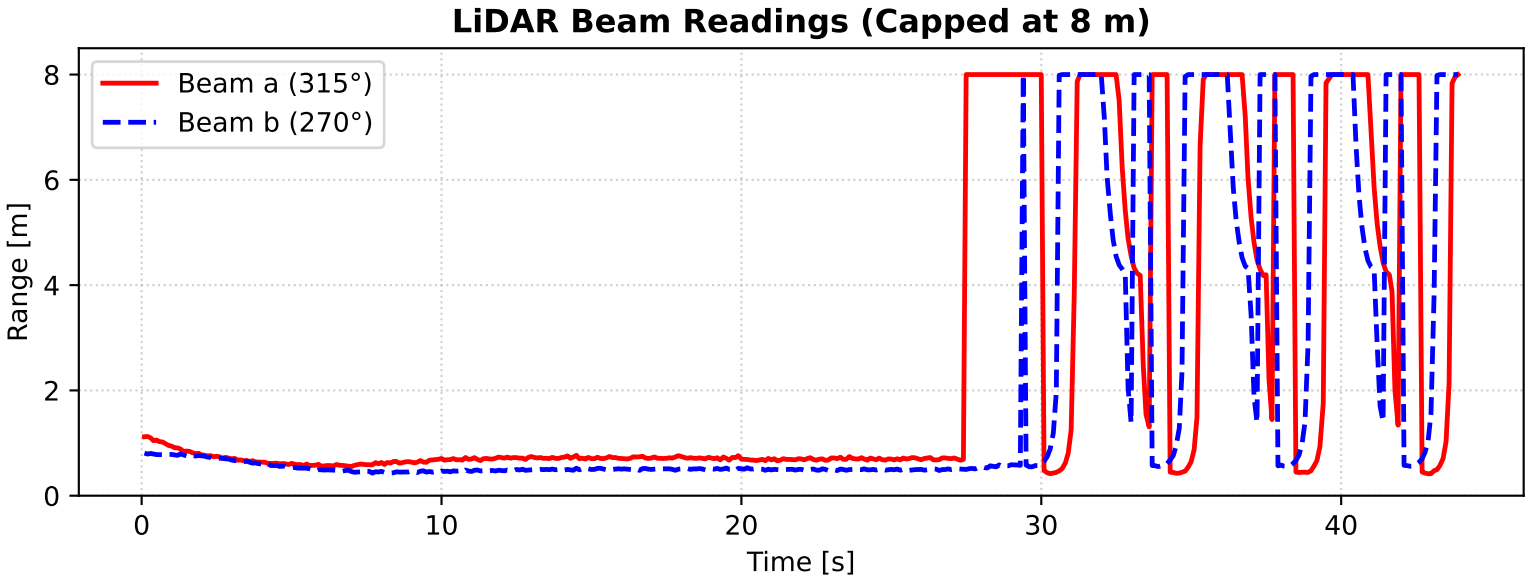
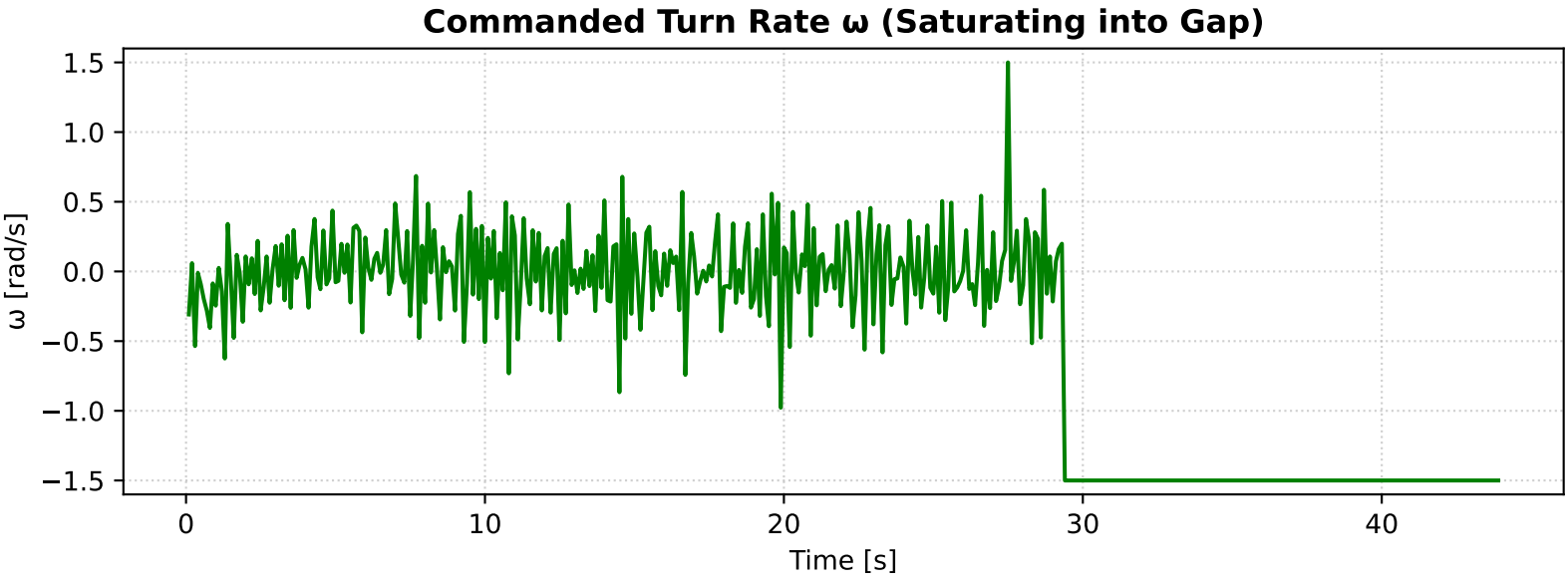

Problem 7(e): Trajectory Turning into the Doorway
Robot Trajectory
Solid Wall
Doorway Gap (1 m)
y [m]
x [m]
−4.5
−5.0
−5.5
−6.0
−6.5
−7.0
−4
−3
−2
−1
0
1
2
3
4


LiDAR Beam Readings (Capped at 8 m)
Beam a (315°)
Beam b (270°)
Range [m]
Time [s]
0
2
4
6
8
0
10
20
30
40


Commanded Turn Rate ω (Saturating into Gap)
ω [rad/s]
Time [s]
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
10
20
30
40